

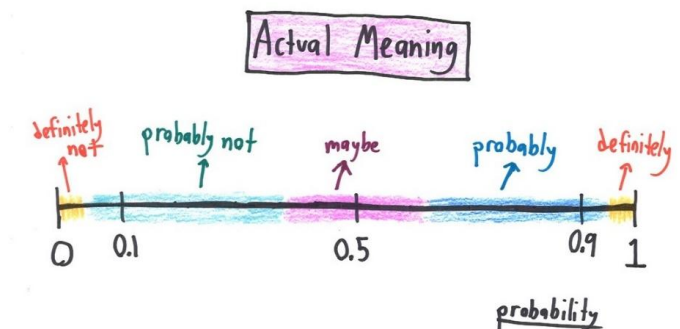


EE 354/CE 361/MATH 310 L2 section (Fall 2025)

Introduction to Probability and Statistics -

aamir hasan

Week 16 - Lecture No 30 – December 03, 2025



Week 16 - Lecture No 30 – December 03, 2025



The Last Lecture

اب اس کے شہر میں ٹھہریں کہ کوچ کر جائیں
فراز آؤ ستارے سفر کے دیکھتے ہیں

Announcements!

1. All lectures posted.
2. Assignment No 3 – by December 05, 2025
3. Quiz No 10 – December 03, 2025
4. End-term exam – Thursday, December 11, 2025 (@Tariq Rafi Hall)

Course Code	Section	Course Title	Date of Examination	Time Start	Time End	Room No.	Enrollment	Name of Instructor(s)
EE/CE 354/361	L1 L2 L3	Introduction to Probability and Statistics	Thursday, December 11, 2025	8:30a	10:30a	E-220	72	Aamir Hasan Tabsheer Ali Askari

Agenda for today

1. Unit 5 – Continuous Random Variables

➡ ➤ Bayes Rule ($X \rightarrow Y$)

1. Discrete $\rightarrow$ Discrete
2. Discrete $\rightarrow$ Continuous
3. Continuous $\rightarrow$ Discrete
4. Continuous $\rightarrow$ Continuous

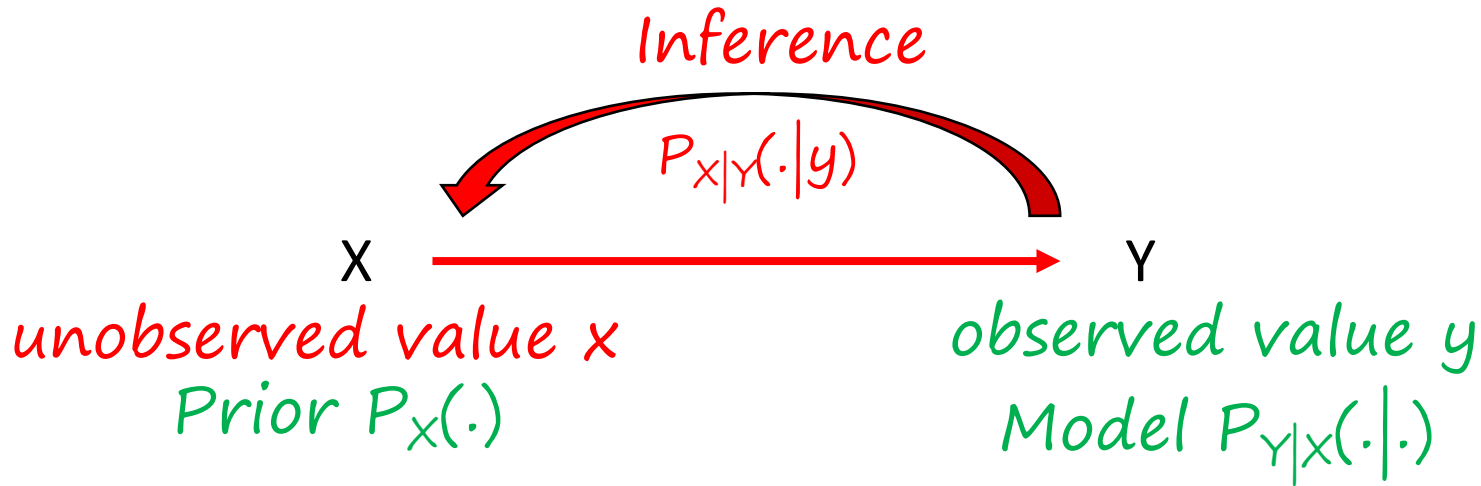
Instructions, please!

1. End-term exam – Thursday, December 11, 2025 (@Tariq Rafi Hall)

Course Code	Section	Course Title	Date of Examination	Time Start	Time End	Room No.	Enrollment	Name of Instructor(s)
EE/CE 354/361	L1 L2 L3	Introduction to Probability and Statistics	Thursday, December 11, 2025	8:30a	10:30a	E-220	72	Aamir Hasan Tabsheer Ali Askari

- 1. Arrive 10 minutes before the start of the exam.**
2. Mobile phones and other electronic devices must be turned off and put away before the start of the exam.
- 3. You are allowed to use a scientific calculator (Not the one on your smart phone. Sharing of calculators is not allowed as well).**
4. No food or drink, other than water, is allowed in the examination room.
5. You are not allowed to communicate with other candidates.
6. You are not allowed to leave your seat unless granted permission by an invigilator.
7. Assume where necessary and justify your assumptions.
8. You are required to fill all answers in the provided answer booklet.
- 9. You are allowed to bring a cheat sheet. It must be a handwritten A4 sheet of paper. You may write on both sides of the sheet. Photocopied cheat sheets are not allowed.**
10. The standard normal table has been provided on the last page.

The Bayes Rule – a theme with variations



$$p_{X,Y}(x, y) = p_Y(y) p_{X|Y}(x|y)$$

$$p_{X,Y}(x, y) = p_X(x) p_{Y|X}(y|x)$$

$$p_{X|Y}(x|y) = \frac{p_X(x) p_{Y|X}(y|x)}{p_Y(y)}$$

$$p_Y(y) = \sum_x p_X(x) p_{Y|X}(y|x)$$

$$f_{X,Y}(x, y) = f_Y(y) f_{X|Y}(x|y)$$

$$f_{X,Y}(x, y) = f_X(x) f_{Y|X}(y|x)$$

$$f_{X|Y}(x|y) = \frac{f_X(x) f_{Y|X}(y|x)}{f_Y(y)}$$

$$f_Y(y) = \int f_X(x) f_{Y|X}(y|x) dx$$

The Bayes Rule – one discrete and one continuous RV

- K: discrete
- Y: continuous

$$P(K=k, y \leq Y \leq y+\delta) = \delta > 0, \delta \approx 0$$

$$P(K=k) \cdot P(y \leq Y \leq y+\delta \mid K=k) \approx P_K(k) f_{Y|K}(y|k) \delta$$

$$P(K=k \mid y \leq Y \leq y+\delta) \cdot P(y \leq Y \leq y+\delta) \approx f_Y(y) \delta P_{K|Y}(k|y)$$

$$P_{K|Y}(k|y) = \frac{P_K(k) f_{Y|K}(y|k)}{f_Y(y)}$$

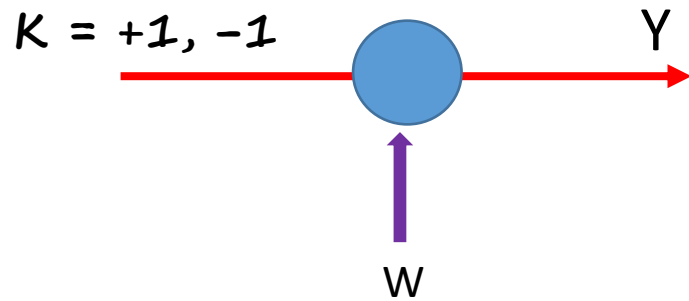
$$f_{Y|K}(y|k) = \frac{f_Y(y) P_{K|Y}(k|y)}{P_K(k)}$$

$$f_Y(y) = \sum_{k'} p_K(k') f_{Y|K}(y|k')$$

$$p_K(k) = \int f_Y(y') p_{K|Y}(k|y') dy'$$

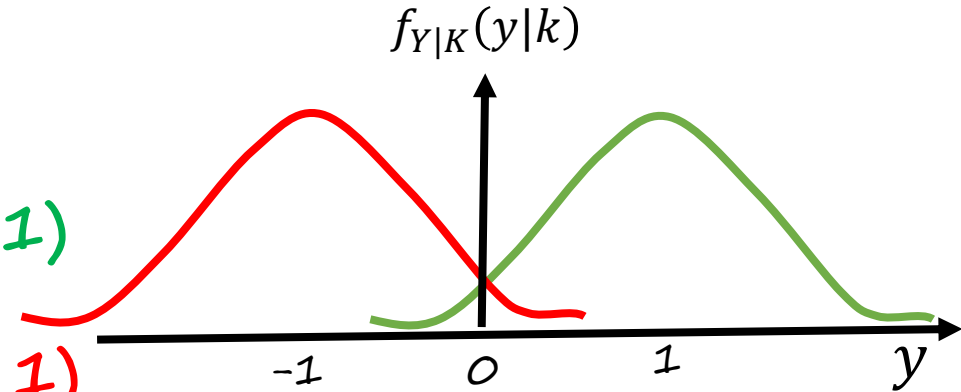
The Bayes Rule – discrete unknown and continuous measurement

- Unknown K : equally likely to be $+1, -1$
- measurement Y : $Y = K + W$ $W \sim N(0, 1)$



$$Y | K = +1 \sim N(1, 1)$$

$$Y | K = -1 \sim N(-1, 1)$$



- Probability that $K = +1$, given that $Y = y$? $P_{K|Y}(1|y)$

- $P_K(k) = 1 / 2$

$$K = +1, -1$$

- $f_{Y|K}(y|k) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(y-k)^2}$

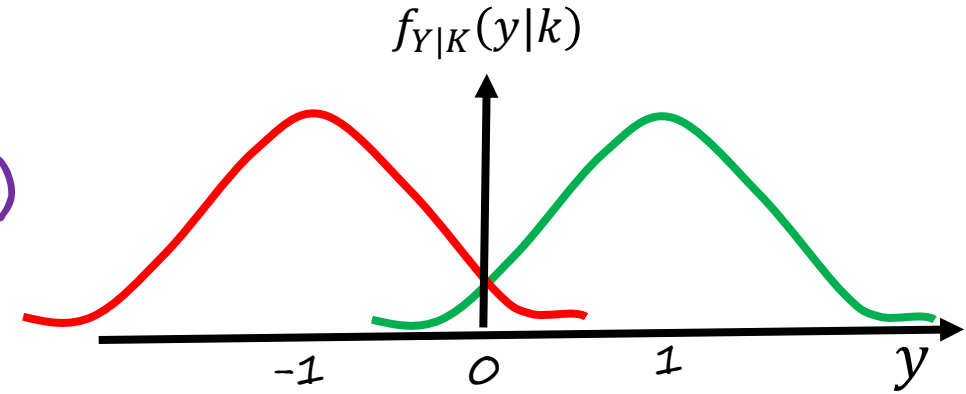
$$P_{K|Y}(k|y) = \frac{P_K(k) f_{Y|K}(y|k)}{f_Y(y)}$$

$$f_Y(y) = \sum_{k'} p_K(k') f_{Y|K}(y|k')$$

The Bayes Rule – discrete unknown and continuous measurement

- Unknown K : equally likely to be $+1, -1$
- measurement Y : $Y = K + W$
- Probability that $K = 1$, given that $Y = y$?

$$W \sim N(0, 1)$$



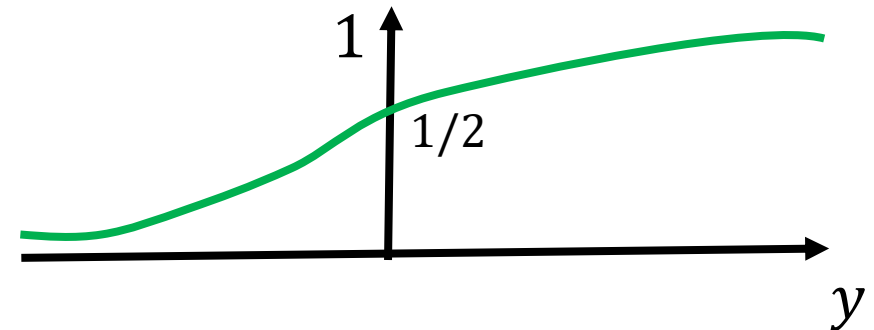
$$P_{K|Y}(k|y) = \frac{P_K(k) f_{Y|K}(y|k)}{f_Y(y)}$$

$$f_Y(y) = \frac{1}{2\sqrt{2\pi}} e^{-\frac{1}{2}(y+1)^2} + \frac{1}{2\sqrt{2\pi}} e^{-\frac{1}{2}(y-1)^2}$$

$$f_Y(y) = \sum_{k'} p_K(k') f_{Y|K}(y|k')$$

$$P_{K|Y}(+1|y) = \frac{1}{1 + e^{-2y}}$$

Interpretation???



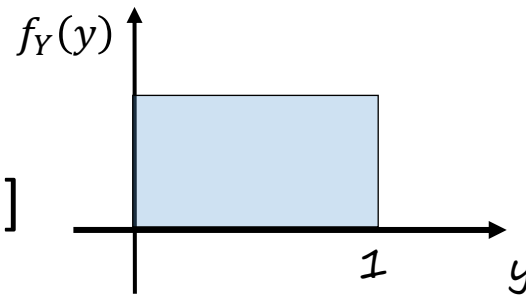
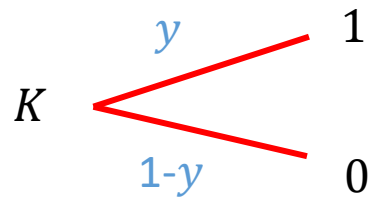


Today - Example

The Bayes Rule – continuous unknown and discrete measurement

The Bayes Rule – continuous unknown and discrete measurement

- K: Bernoulli with parameter Y



- unknown Y: uniform on [0, 1]

- Distribution of Y given that K = 1? $f_{Y|K}(y|1)$

$$f_Y(y) = \begin{cases} 1, & y \in [0, 1] \\ 0, & \text{otherwise} \end{cases}$$

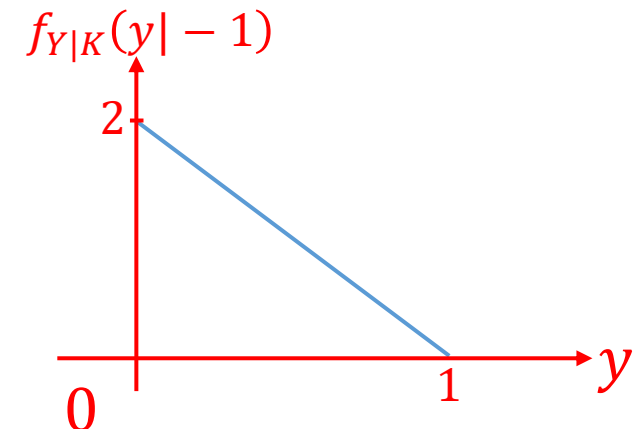
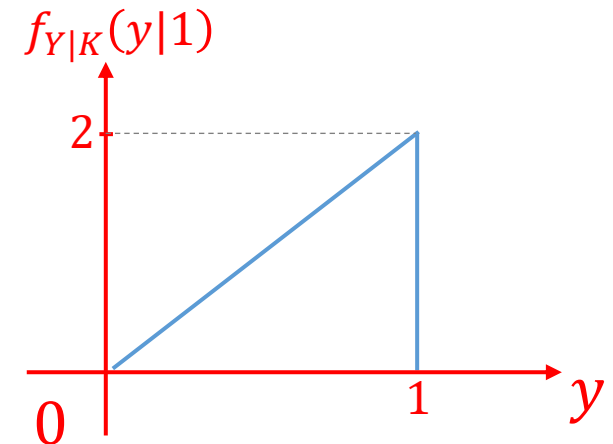
$$p_{K|Y}(1|y) = y$$

$$p_K(1) = \int_0^1 1 \cdot y \, dy = 1/2$$

$$f_{Y|K}(y|1) = \frac{1 \cdot y}{1/2} = 2y, \quad y \in [0, 1]$$

$$f_{Y|K}(y|k) = \frac{f_Y(y) P_{K|Y}(k|y)}{P_K(k)}$$

$$p_K(k) = \int f_Y(y') p_{K|Y}(k|y') \, dy'$$



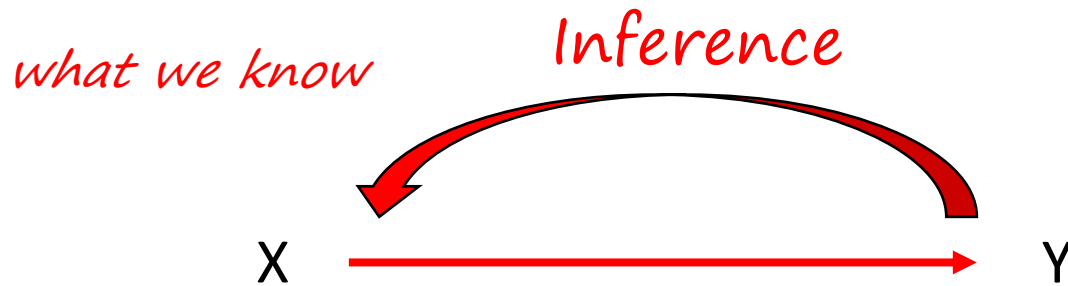
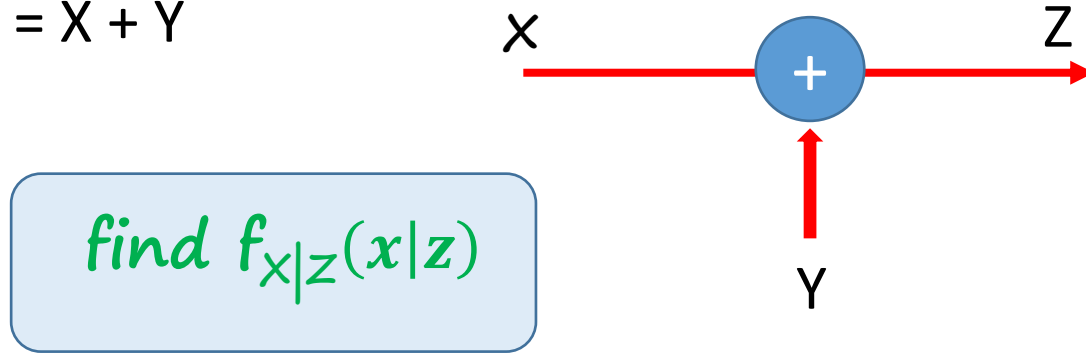


Example

The Bayes Rule – continuous unknown and continuous measurement

The Bayes Rule – Example: Two continuous R.Vs

- X, Y independent continuous R.Vs (i.i.d)
- $Z = X + Y$



- In this case, we observe Z and infer X

$$f_{X|Y}(x|y) = \frac{f_X(x)f_{Y|X}(y|x)}{f_Y(y)}$$



$$f_{X|Z}(x|z) = \frac{f_X(x)f_{Z|X}(z|x)}{f_Z(z)}$$

The Bayes Rule – Example: Two continuous R.Vs

- X, Y independent continuous R.Vs (i.i.d)
- $Z = X + Y$

find $f_{X|Z}(x|z)$

$$f_{X|Z}(x|z) = \frac{f_X(x)f_{Z|X}(z|x)}{f_Z(z)}$$

lets determine the model $f_{Z|X}(z|x)$ first

$$\text{we know } Z = X + Y \Rightarrow Z = x + Y$$

$$\Rightarrow Y = Z - x$$

$$\text{So, } f_{Z|X}(z|x) = f_Y(z-x)$$

Why?

$$\text{Since, } Z = x + Y$$

$$z = x + (z - x)$$

lets do it more formally!

The Bayes Rule – Example: Two continuous R.Vs

lets do it more formally!

$$\begin{aligned}P(Z \leq z \mid X=x) &= P(X+Y \leq z \mid X=x) \\&= P(x+Y \leq z \mid X=x) \\&= P(x+Y \leq z) \\&= P(Y \leq z - x)\end{aligned}$$

$$F_{Z|X}(z|x) = F_Y(z - x)$$

$$dF_Y(z - x)/dy = f_Y(z-x); \frac{dy}{dz}=1$$

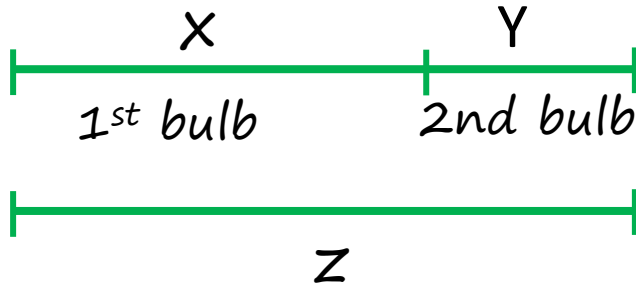
$$\text{so, } f_{Z|X}(z|x) = f_Y(z-x)$$

$$f_{X|Z}(x|z) = \frac{f_X(x)f_{Z|X}(z|x)}{f_Z(z)}$$

$$f_{Z|X}(z|x) = f_Y(z-x)$$

The Bayes Rule – Example: Two continuous R.Vs

- Now assume X, Y are exponential R.Vs with parameter λ (i.i.d)



$$f_{X|Z}(x|z) = \frac{f_X(x)f_{Z|X}(z|x)}{f_Z(z)}$$
$$f_{Z|X}(z|x) = f_Y(z-x)$$

You observe Z to make an inference about X (1st light bulb)

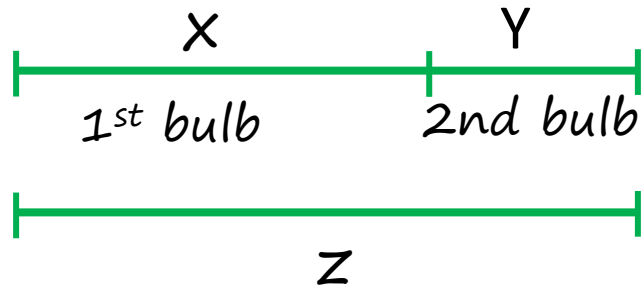
$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

$$f_Y(y) = \lambda e^{-\lambda y}, \quad y \geq 0$$

fix some $z \geq 0$

$$f_{X|Z}(x|z) = \frac{\lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)}}{f_Z(z)}$$
$$= \frac{1}{f_Z(z)} \lambda^2 e^{-\lambda z} \quad 0 \leq x \leq z$$

The Bayes Rule – Example: Two continuous R.Vs



for some z ; $f_Z(z)$ is a scalar quantity

$$f_{X|Z}(x|z) = \frac{1}{f_Z(z)} \lambda^2 e^{-\lambda z} \quad 0 \leq x \leq z$$

$$f_{X|Z}(x|z) = \text{uniform on } [0, z]$$

Why?

Think

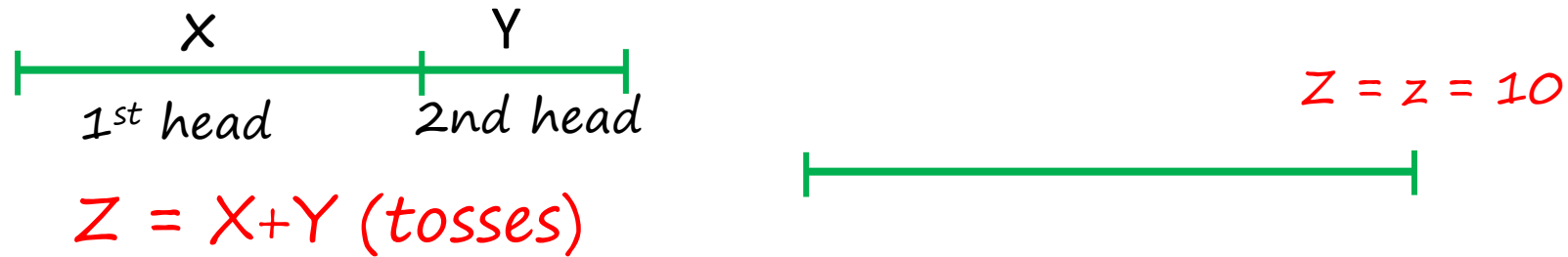


Discuss



Answer

The Bayes Rule – Example: Two independent tosses of coin



Given that 2nd head is on the tenth toss ($z = 10$), the first toss could be anywhere between 1 to 9

$$P(X=1|Z=10) = P(X=1, Y=9) = p \cdot (1-p)^8 p = p^2(1-p)^8$$

$$P(X=2|Z=10) = P(X=2, Y=8) = (1-p)p \cdot (1-p)^7 p = p^2(1-p)^8$$

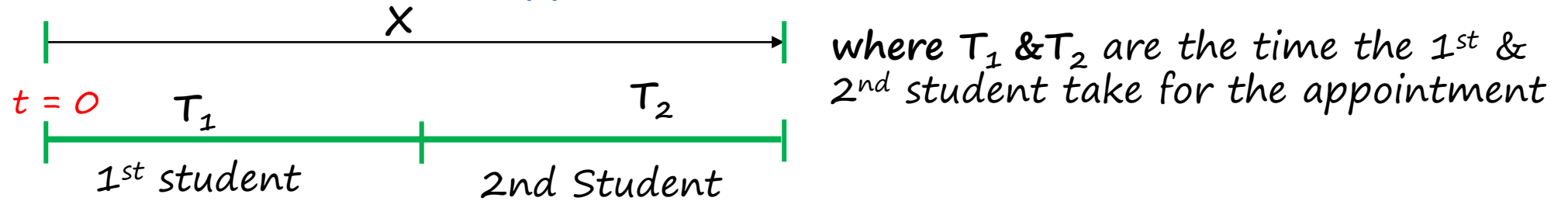
⋮

$$P(X=9|Z=10) = P(X=9, Y=1) = (1-p)^8 p \cdot p = p^2(1-p)^8$$

$P_{X|Z}(x|10)$ is uniformly distributed $X|Z \sim U[1, 9]$

Home Assignment

Example: Absent minded Professor – 2 appointments w/ two students at the same time



$$T_1 \text{ \& } T_2 \sim \exp(\lambda), \quad P(T_1 < t) = 1 - e^{-\lambda t}, \text{ for } t > 0 \quad P(T_2 < t) = 1 - e^{-\lambda t}, \text{ for } t > 0$$
$$E[T_1] = E[T_2] = 1/\lambda$$

Given that 2nd student arrives exactly after 5 minutes of the arrival of the 1st student

Comment on the $E[X]$ – where X is the total duration of the two appointments & for both students $\lambda = 1/30$ mins

Case 1 – condition $T_1 < 5$ mins $E[X \mid T_1 < 5] = 5 + E[T_2] = 35$ mins

Case 2 – condition $T_1 > 5$ mins $E[X \mid T_1 > 5] = 5 + E[T_1] + E[T_2] = 5 + 30 + 30 = 65$ mins

$$E[X] = P(T_1 < t) \cdot E[X \mid T_1 < 5] + P(T_1 > t) \cdot E[X \mid T_1 > 5]$$

Example Bayes Rule – Continuous – Discrete RVs

Example pertains to inferring a continuous random variable Q , from a discrete observation of a random variable X . Let Q be a continuous random variable with PDF as follows:

$$f_Q(q) = \begin{cases} 6q(1 - q), & \text{if } 0 \leq q \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Q represents the probability of a success of a Bernoulli random variable X , i.e., $P(X = 1 | Q = q) = q$ & $P(X = 0 | Q = q) = (1 - q)$. Find $f_{Q|X}(q|x)$ for $x \in \{0, 1\}$, for all possible values of q . You may use following version of Bayes' rule:

$$f_{Q|X}(q|x) = \frac{f_Q(q)p_{X|Q}(x|q)}{p_X(x)}$$

$$\text{Given } f_{\theta|X}(u|x) = \frac{f_{\theta}(u) \cdot P_{X|\theta}(x|u)}{P_X(x)}$$

$$\text{where } P_X(x) = \int_0^1 f_{\theta}(u) \cdot P_{X|\theta}(x|u) du \quad \text{for } x=0,1 \text{ \& } u \in [0,1]$$

for $x=0$

$$f_{\theta|X}(u|0) = \frac{6u(1-u) \cdot (1-u)}{\int_0^1 6u(1-u)(1-u) du} \quad ; \quad P_{X|\theta}(0|u) = (1-u)$$

$$= \frac{6u(1-u)(1-u)}{6 \int_0^1 u + u^3 - 2u^2 du}$$

$$= \frac{6u(1-u)(1-u)}{6 \left[u^2/2 + u^4/4 - 2u^3/3 \right]_0^1} = \frac{6u(1-u)(1-u)}{1/2}$$

$$= 12u(1-u)^2$$

$$0 \leq u \leq 1$$

$$f_{Q|X}(q|x) = \frac{f_Q(q)p_{X|Q}(x|q)}{p_X(x)}$$

$$P(X=1|Q=q) = q \text{ \& } P(X=0|Q=q) = (1-q).$$

for $x=1$ and $q \in [0,1]$

$$f_{\theta|X}(q|1) = \frac{6q(1-q) \cdot q}{1/2}$$

as $P_{X|Q}(1|q) = q$

$P_X(0) = 1/2 \Rightarrow$ ^{implies} $P_X(1) = 1/2$

$$f_{\theta|X}(q|1) = 12q^2(1-q)$$

$$0 \leq q \leq 1$$

$$f_{Q|X}(q|x) = \frac{f_Q(q)p_{X|Q}(x|q)}{p_X(x)}$$

$$P(X=1|Q=q) = q \text{ \& } P(X=0|Q=q) = (1-q).$$

Congrats!

End of Unit 5!

Practice Problem

Question 2 - [6 points]: A wire has length X , an exponential random variable with parameter $\lambda = 1/\pi$ cm. The wire is cut to make rings of diameter 1 cm. Find the PMF $P_N(n)$, the probability for the number of complete rings N produced by each length of wire.

Hint – use CDF $F_X(x)$

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

- Continuous random variables:

$$F_X(x) = P(X \leq x) = \int_{-\infty}^x f_X(t) dt$$

$$F_X(x) = \begin{cases} 1 - e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Question 2 - [6 points]: A wire has length X , an exponential random variable with parameter $\lambda = 1/\pi$ cm. The wire is cut to make rings of diameter 1 cm. Find the PMF $P_N(n)$, the probability for the number of complete rings N produced by each length of wire.

Hint – use CDF $F_X(x)$

$$P(N=0) = F_X(\pi)$$

$$P(N=1) = F_X(2\pi) - F_X(\pi)$$

$\vdots$

$$P(N=n) = F_X((n+1)\pi) - F_X(n\pi)$$

$$= 1 - e^{-\lambda(n+1)\pi} - (1 - e^{-\lambda n\pi}) \quad \text{--- 2}$$

$$= 1 - e^{-\lambda(n+1)\pi} + e^{-\lambda n\pi}$$

$$= e^{-\lambda n\pi} (-e^{-\lambda\pi} + 1)$$

$$= e^{-\frac{n\pi}{\pi}} (1 - e^{-1})$$

$$\boxed{P_N(n) = e^{-n} (1 - e^{-1})} \quad \text{--- 2}$$

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

• Continuous random variables:

$$F_X(x) = P(X \leq x) = \int_{-\infty}^x f_X(t) dt$$

$$F_X(x) = \begin{cases} 1 - e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Home assignment

CLO No 4 - Construct probability distributions from given data for discrete and continuous random variables.

Problem 3 (20 points): Find the CDF and PDF for the Random Variable Z and **specify the range of values where they are applicable:**

Part a (10 points): A Random Variable Z is defined as $Z = \min(X_1, X_2, \dots, X_n)$, where X_i 's are independent random variables with X_i having an exponential distribution with rate λ_i for $i = 1, \dots, n$.

Part b (10 points): Let X and Y be independent random variables that are uniformly distributed with $X \sim U[0, 1]$ and $Y \sim U[0, 2]$. The random variable $Z = \max(X, Y)$.

Problem 3 (20 points): Find the CDF and PDF for the Random Variable Z and **specify the range of values where they are applicable:**

Part a (10 points): A Random Variable Z is defined as $Z = \min(X_1, X_2, \dots, X_n)$, where X_i 's are independent random variables with X_i having an exponential distribution with rate λ_i for $i = 1, \dots, n$.

X_i are exponentially distributed, and so the PDF of each is:

$$f_{X_i}(x_i) = \begin{cases} \lambda_i e^{-\lambda_i x_i} & \text{if } x_i > 0 \\ 0 & \text{otherwise.} \end{cases}$$

Now, we need to find out the CDF of Z . Since X_i cannot take values below 0, it is clear that $F_Z(z) = 0$ for all $z < 0$. For the case $z \geq 0$,

$$\begin{aligned} F_Z(z) &= P(Z \leq z) \\ &= P(\min(X_1, \dots, X_n) \leq z) \\ &= 1 - P(\min(X_1, \dots, X_n) > z) \\ &= 1 - P(X_1 > z, X_2 > z, \dots, X_n > z) \end{aligned}$$

Since X_i are independent,

$$F_Z(z) = 1 - P(X_1 > z) P(X_2 > z) \cdots P(X_n > z)$$

Let's find out the probability that $X_i > z$.

$$P(X_i > z) = \int_z^\infty \lambda_i e^{-\lambda_i x_i} dx_i = e^{-\lambda_i z}$$

Thus,

$$\begin{aligned} F_Z(z) &= 1 - e^{-\lambda_1 z} e^{-\lambda_2 z} \cdots e^{-\lambda_n z} \\ &= 1 - e^{-(\lambda_1 + \cdots + \lambda_n)z} \end{aligned}$$

Thus,

$$F_Z(z) = \begin{cases} 0 & \text{if } z < 0 \\ 1 - e^{-(\lambda_1 + \cdots + \lambda_n)z} & \text{if } z \geq 0. \end{cases}$$

Differentiating this, we can obtain PDF for Z :

$$f_Z(z) = \begin{cases} (\lambda_1 + \cdots + \lambda_n) e^{-(\lambda_1 + \cdots + \lambda_n)z} & \text{if } z \geq 0 \\ 0 & \text{if } z < 0 \end{cases}$$

Problem 3 (20 points): Find the CDF and PDF for the Random Variable Z and **specify the range of values where they are applicable:**

Part b (10 points): Let X and Y be independent random variables that are uniformly distributed with $X \sim U[0, 1]$ and $Y \sim U[0, 2]$. The random variable $Z = \max(X, Y)$.

The PDFs of two variables are:

$$f_X(x) = \begin{cases} 1 & \text{if } x \in [0, 1] \\ 0 & \text{otherwise.} \end{cases}$$

$$f_Y(y) = \begin{cases} \frac{1}{2} & \text{if } y \in [0, 2] \\ 0 & \text{otherwise.} \end{cases}$$

$$\begin{aligned} F_Z(z) &= P(Z \leq z) \\ &= P(\max(X, Y) \leq z) \\ &= P(X \leq z, Y \leq z) \end{aligned}$$

Problem 3 (20 points): Find the CDF and PDF for the Random Variable Z and **specify the range of values where they are applicable:**

Part b (10 points): Let X and Y be independent random variables that are uniformly distributed with $X \sim U[0, 1]$ and $Y \sim U[0, 2]$. The random variable $Z = \max(X, Y)$.

The PDFs of two variables are:

$$f_X(x) = \begin{cases} 1 & \text{if } x \in [0, 1] \\ 0 & \text{otherwise.} \end{cases}$$

$$f_Y(y) = \begin{cases} \frac{1}{2} & \text{if } y \in [0, 2] \\ 0 & \text{otherwise.} \end{cases}$$

$$\begin{aligned} F_Z(z) &= P(Z \leq z) \\ &= P(\max(X, Y) \leq z) \\ &= P(X \leq z, Y \leq z) \end{aligned}$$

By independence of X and Y ,

$$\begin{aligned} &= P(X \leq z) P(Y \leq z) \\ &= \begin{cases} 0 & \text{if } z < 0 \\ z \times \frac{1}{2} & \text{if } 0 \leq z \leq 1 \\ 1 \times \frac{1}{2} & \text{if } 1 \leq z \leq 2 \\ 1 \times 1 & \text{if } z > 2 \end{cases} \end{aligned}$$